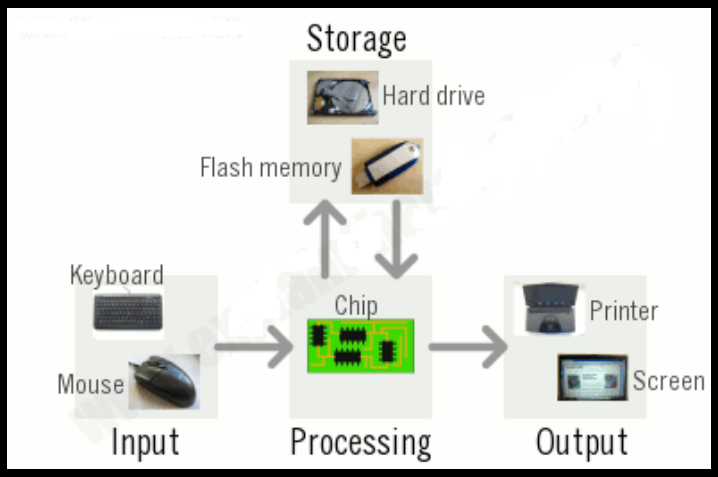
How A Computer Works

A **computer** is an electronic machine that processes information

1) The Computer accepts data from one of the Input devices (keyboard, mouse, etc.)

2) Information is put into memory for temporary storage (RAM) for processing

3) The CPU processes the info and decides what to do with it - usually sent to an OUTPUTdevice (monitor, printer, etc.)



The Motherboard :

The motherboard is the most important piece of hardware in your computer. All the components are connected though the Motherboard. The components that control your computer (CPU, RAM, HDD, etc.) are all directly on the Motherboard.

The CPU :

- This is the "brains" of the computer. The CPU or "Central processing unit", like all computer chips, is made up of silicon with tiny electronic circuits etched on it. Data is processed by passing electricity though the circuits (silicon will not conduct electricity).
- The CPU speed is a major factor in determining how fast your computer will operate. CPU speed is measured in Mhz (Mega hertz) or millions of cycles per second.

- The CPU has 2 main functions
  1) CONTROL **UNIT** controls how instructions/data is handled and decides what goes to the ALU
  2)ALU (Arithmetic Logic Unit) which is responsible for all arithmetic and logical operations and processing

Operating System :

- The computer's operating system is the software that controls the overall activity of the computer. It ensures that all parts work together smoothly and efficiently. It manages the hardware and assists with software operations.
  ex : Windows, OS (mac), LINUX

BIOS :

- The BIOS chip is located on the motherboard. The BIOS is responsible for "booting" the computer when the power is turned on.
- During the boot process the BIOS checks the processor, RAM, Drives, etc. to ensure correct operation. If all is OK the Operating System is loaded
- The BIOS also loads all the "driver" for each piece of hardware.

How the Computer Communicates :

- Data is transmitted from one part to another using a collection of wires called a "**BUS**". All the internal components are connected using Buses. A Bus consists of 2 parts :
  1) CONTROL Bus
  2) ADDRESS Bus

  The Data bus transfers actual data whereas the Address Bus will determine where the data will go.

- A "PORT" is a connection point where external devices (mouse, keyboard, monitor, etc.) are connected to the computer (motherboard)

Questions and Exercises :

1. List 5 input devices.Keyboard,Mouse,Scanner,Microphone,Webcam
2. List 5 output devices.Monitor,Printer,Speaker,Headphones,Projector
3. List 3 input and output devices.Touchscreen,Network card,Modem
4. Where does data coming from an input device get sent to before it is processed?It gets sent to the RAM (Random Access Memory).
5. What device in the computer processes all input data?The CPU (Central Processing Unit).
6. What is the 'brains of the computer'?The CPU (Central Processing Unit).
7. What is the unit of measuring the speed of a CPU?Gigahertz (GHz) or Megahertz (MHz).
8. What are the two main functions of a CPU? Processing data and instructions (arithmetic and logical operations).Controlling the operation of other computer components.
9. What role does the OS play in the operation of a computer?The Operating System (OS) manages computer hardware and software resources, provides an interface for users to interact with the computer, and ensures programs run smoothly.
10. What is the BIOS responsible for in a computer?BIOS (Basic Input/Output System) is responsible for initializing and testing hardware components during startup (POST - Power - On Self - Test) and loading the operating system.
11. What is a computer BUS?A computer bus is a communication pathway that transfers data, power, and control signals between different computer components (like CPU, memory, and peripherals).
12. What two main types of BUS exist in the computer?
    Data Bus
    Address Bus

Below is a list of some common computer ports. Next to each port add an image and a short description of what types of devices they are commonly attached to. When you are done, go with your group and select one of the computers in the class. Get a label from your teacher. Clean the top of the computer. Stick the label on top of the

computer and print all group member full names on it. Gather all necessary peripheral devices, cables etc. and set up carefully and test that your computer works. This computer and its peripheral devices now belong to you for the duration of this course. You need to take care of it and make sure the workspace around it is always neat and tidy.

- Serial port Usually a 9-pin or 25-pin D-shaped connector. Used for old modems, industrial equipment, and some legacy devices.
- Parallel port A 25-pin connector, wider than serial port. Used for old printers, scanners (legacy).
- USB port Rectangular (Type A/C). For keyboards, mice, drives, phones, printers.
- PS/2 port Round, purple/green. For legacy keyboards, mice.
- VGA port Blue, 15-pin. For old monitors, projectors.
- Modem port RJ-11/25-pin. For old dial-up modems.
- FireWire Port 6/9-pin rectangular. For external drives, video cameras.
- Sockets Standard power. For computers, peripherals (power).
- Infrared Port Recessed window. For old wireless data transfer (phones, printers).
- Game Port 15-pin. For legacy joysticks, gamepads.
- Digital Video Interface(DVI) Port White rectangular. For monitors, projectors (digital video).
- Ethernet Port RJ-45. For network cables, routers (internet/LAN).

After completing the ports work from above, carefully unplug all peripheral devices, including the power cable from your computer. Take a picture of all front and back panel ports. Share the image with yourself and use any tool (https://markuphero.com/) you are comfortable with, label the ports. Embed that image here: